

Program: Diploma in Fire Technology and Safety	
Course Code: 6462C	Course Title: Transportation Safety
Semester: 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To understand the key concepts of railway engineering and its safety aspects.
- To learn the fundamentals of highway design, including road patterns and alignment.
- To explore traffic engineering principles and how to ensure road safety.
- To gain insight into harbor and dock engineering, focusing on port design and safety in water transportation.

Course Outcomes

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the basic concepts of railway engineering, operations, and safety.	15	Understanding
CO2	Learn the basic principles of highway engineering, road patterns, and alignment considerations.	14	Understanding
CO3	Understand traffic engineering concepts and methods for improving road safety.	14	Understanding
CO4	Learn about the design of harbors and docks, and understand the safety features of water transportation systems.	15	Understanding
	Series Test	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the basic concepts of railway engineering, operations, and safety.		
M1.01	Introduction to railway engineering and permanent way design..	3	Understanding
M1.02	Basics of railway curves, super-elevation, and transition curves.	4	Understanding
M1.03	Railway operations: points, crossings, and signaling systems.	4	Understanding
M1.04	Safety measures in railway operations, including accident prevention.	4	Understanding
Contents: Railway track components and permanent way design. Types of railway curves and their effects on safety. Signaling systems and their role in railway operations. Railway accident prevention and safety protocols..			
CO2	Learn the basic principles of highway engineering, road patterns, and alignment considerations.		
M2.01	Classification of highways and types of road patterns.	3	Understanding

M2.02	Basic concepts in road alignment and geometric design.	4	Understanding
M2.03	Design considerations for road safety, including sight distances.	4	Understanding
M2.04	Factors affecting the safe design of roads and intersections.	3	Understanding
	Series Test I	1	
Contents: Road classification and typical urban road patterns. Geometric design principles for road alignment. Safety considerations in road design, including sight distances. Factors influencing road safety and intersection design.			
CO3	Understand traffic engineering concepts and methods for improving road safety.		
M3.01	Traffic flow characteristics and road safety.	3	Understanding
M3.02	Traffic control devices: signals, road markings, and traffic islands.	4	Understanding
M3.03	Principles of highway lighting and its role in road safety.	4	Understanding
M3.04	Road accident causes, prevention methods, and safety measures.	3	Understanding
Contents: Principles of traffic flow and road safety. Traffic signals, markings, and their role in managing traffic. The importance of proper highway lighting for night driving. Methods for preventing and reducing road accidents.			

CO4	Learn about the design of harbors and docks, and understand the safety features of water transportation systems.		
M4.01	Basics of water transportation and harbor classifications.	4	Understanding
M4.02	Port design: layout, accessibility, and functionality.	4	Understanding
M4.03	Key facilities in ports: docking, repair, and unloading.	3	Understanding
M4.04	Safety measures for harbor and dock operations	4	Understanding
	Series Test II	1	
Contents: Water transportation systems and harbor classifications. Design and layout considerations for ports. Essential facilities in ports: docking, repair, and unloading. Safety protocols in harbor and dock operations.			

Text/Reference:

T/R	Book Title/Author
R1	Khanna, S.K., & Justo, C.E.G., "Highway Engineering," Nem Chand & Bros, 11th Edition, 2014.
R2	Chowdhury, S.K., "Railway Engineering," Oxford University Press, 3rd Edition, 2018.
R3	O'Flaherty, C.A., "Traffic Engineering," Butterworth-Heinemann, 3rd Edition, 2008.
R4	Iremonger, W., "Harbour and Dock Engineering," McGraw-Hill, 2nd Edition, 1999.